Diagonality: The Best of Both Worlds

PR / FAQ

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Press Release

Football's most-discussed tactical idea finally has a number, and a real-time map.

Diagonality, the preference for oblique passes, carries and runs over straight ones, has been argued about for decades but never measured. A new framework built on 3D skeleton tracking validates the theory against 6,923 on-ball actions and turns it into the Diagonal Opportunity Surface: a live, orientation-aware map of where, when and against which defender to attack diagonally.

Frankfurt, 16 May 2026. Coaches have a name for the smartest ball a team can play, the diagonal, and a theory for why it works: Spielverlagerung's 2025 essay calls it "the best of both worlds", a pass that carries the progression of a vertical ball and the safety of a horizontal one. The essay also hedges every claim it makes, bracketing the word "(assumed)", because diagonality is at bottom a statement about *body orientation*, and tracking data has only ever recorded *positions*. The idea could be admired but not tested, and certainly not turned into a tool.

Diagonality closes that gap. Using TRACAB 3D skeleton tracking from five Bundesliga 2025–26 matches (21 keypoints per player at 50 Hz, the first dataset that measures the real head, shoulder and hip orientation of every footballer), the project does two things. First, it **tests the theory**: the essay's argument is distilled into a perceptual premise and three falsifiable hypotheses, tested against 6,923 passes, carries and take-ons with metrics that owe nothing to the new model. The claims hold. Diagonal actions are the efficient compromise between progression and safety, they disrupt the defensive block on the lateral axis a vertical pass leaves untouched, and they hand the receiver a body already turned toward goal.

Second, it **turns the validated theory into a tool**: the **Diagonal Opportunity Surface (DOS)**. DOS is a real-time pitch map that fuses what every defender can see, how each is oriented, and who controls space, into a single signal: the extra control an attacker buys by delivering the ball diagonally rather than straight, into the space behind a defender who cannot see it. A high-DOS zone is not a black box: it decomposes into the three causes the tactical theory names: a defender's blind spot, the orientation delay in reaching it, and the directional benefit of arriving oblique.

"Diagonality was the game's most-discussed unquantified idea, and it stayed unquantified for a concrete reason: it is a claim about body orientation, and tracking never recorded orientation," said Jaime Oriol, who built the framework. "With 3D skeleton data we could finally measure it. We tested the theory and it held; then we turned it into a map a coach can read frame by frame. We turned a sentence into a number on a pitch."

For a coaching or analysis desk, DOS is immediately usable. Aggregated per opponent, it is a scouting map of where a specific team's defenders habitually turn away from danger. Aggregated per player, it is a profile, and it passes the analysts' informal "Messi test": ranked by the total diagonal opportunity they generate, the leaderboard is headed by exactly the players the eye already knows as creators. Gated to what the player on the ball can actually perceive, DOS becomes a real-time decision aid, showing which passes, dribbles and off-ball runs are both available and visible, plus a second amber layer of high-value options the carrier has stopped watching and should turn to re-check.

“The honest problem with ‘play more diagonals’ as an instruction is that nobody could point to where or when,” said a hypothetical performance analyst, the kind of user the framework is built for. “A surface that lights up the blind side, per opponent, turns a slogan into something you can coach on Monday and check on Saturday.”

The framework is released as open source. Every stage runs from the repository root, is memory-safe and deterministic, and is documented for reproduction; no proprietary hackathon data is redistributed. The same pipeline scales without modification from five matches to a full season.

Frequently Asked Questions: External

What exactly is the Diagonal Opportunity Surface?

DOS is a value, computed for every cell of the pitch at every frame, equal to the extra ball control the attacking team would gain by playing into that cell *diagonally* rather than straight. It is built from a per-defender vision model, an orientation-aware pitch-control function and a cognitive scanning gate. The same routine scores a pass, a carry, a take-on or an off-ball run, because all four reduce to an origin and a direction.

How is this different from existing pitch-control models?

Every pitch-control model in the literature treats a player as a disc or an isotropic Gaussian, equally fast and equally aware in all directions. None encodes which way the player is facing. DOS is built on the opposite premise: a defender controls the space in front of them far better than the space behind their shoulder, because their *perception* of space is asymmetric. That asymmetry is the entire mechanism of diagonality, and it requires measured orientation to model.

Why 3D skeleton data and not ordinary tracking?

Ordinary tracking gives positions; body orientation then has to be proxied from the velocity vector, which a player violates at will by running one way while facing another, an error of 30–50°, large enough to drown the signal. Broadcast pose estimation reduces it to about 27°. TRACAB 3D skeleton tracking removes it: the head, shoulder and hip keypoints are measured directly, at 50 Hz.

Did the tactical theory actually hold up?

Yes, on three independent tests. Diagonal actions produce a value-positive outcome more often than forward or sideways actions (the progression–safety trade-off); they carry the highest *combined* longitudinal-plus-lateral disruption of the defensive block; and their receiver needs far less rotation to face goal and sees more team-mates at the moment of reception. Honest nuance: the diagonal does not beat the vertical pass on *depth* disruption: forward actions match it there. The diagonal’s specific edge is the lateral axis, and the combination.

Is DOS biased by hindsight, or does it secretly use outcomes?

No, and this is deliberate. DOS is computed from geometry, the vision model and skeleton orientation alone. It never sees a goal, an xG, an xT or a success label. That independence is what makes the validation fair: when DOS predicts expected-threat gain, it is predicting an outcome it was never shown.

Can it run live during a match?

The model is single-frame and the scanning gate is explicitly real-time: it bounds the surface to what the on-ball player can perceive within a 2.5-second memory window. The current implementation favours clarity and reproducibility over latency optimisation; a production deployment would target the broadcast or bench-tablet frame budget.

Does it work for off-ball runs?

Yes. Run detection is already a solved, commercial product; the framework uses a lightweight detector and applies the same DOS valuation. 2,354 off-ball runs were detected and scored, and the overwhelming majority carry positive DOS: off-ball movement, like on-ball action, is a search for the defender's blind side.

What can a club do with it tomorrow?

Four things: match preparation (a per-opponent map of orientation vulnerabilities), post-match analysis (every action annotated with its DOS and the misalignment it exploited), recruitment (players profiled by the diagonal opportunity they generate), and broadcast (the blind-spot and DOS surfaces as legible overlays).

Frequently Asked Questions: Internal

Why a reach-field pitch-control model instead of standard time-to-intercept?

Because the question is not "who gets there first" but "how well is each player placed, given how they are turned". Each player is modelled as an anisotropic Gaussian whose width follows an orientation-aware delay: a reaction-time term that grows with visual eccentricity plus a change-of-direction penalty, both taken from published biomechanics. A defender with a threat behind their shoulder has a control blob compressed to half-width: a literal hole in the place the theory says the hole should be.

What was the hardest part?

Making a perceptual idea computable without it becoming arbitrary. The vision model, the awareness weighting and the detection delay each had to trace back to a published mechanism rather than a tuned constant, so that a high-DOS cell could be *explained*, not just displayed.

What failed along the way, and what did it teach you?

Several things, and each shaped the final design:

- **Memory.** A single match does not fit in 8 GB of RAM. The naive "load the match" approach failed immediately; every stage was rebuilt around chunked, predicate-pushdown reads over frame ranges. The pipeline is now memory-safe by construction and, as a side effect, trivially scalable.
- **Carries.** Scoring a carry by its start-to-end direction was wrong: a carry serpentines, stops and turns. The fix was to sample the carry at several frames and use the *instantaneous* skeleton velocity as the true direction, taking the diagonal peak.
- **A data quirk.** The DFL event XML does not list carries under their parent possession, which silently broke outcome linkage. It had to be repaired by matching on frame range and team.
- **Visual flicker.** The first real-time DOS render flickered frame to frame. The cause was relative thresholds; the fix was absolute thresholds plus a temporal smoothing pass with a hard reset on possession change.
- **A surprising result.** DOS and the defensive-disruption metric D-Def turned out only weakly (and negatively) correlated. The first instinct was that something was broken. It was not: DOS measures the perceptual *opportunity* at the instant of the action, D-Def the structural *consequence* three seconds later. They are complementary, and the gap between them is itself a coaching signal.

Why the "Messi test", and did it pass?

A metric that claims to capture a footballing quality must rank the players the eye already knows for it, or it is measuring something else. Ranked by total diagonal opportunity generated, the leaderboard is headed by a recognisable creative core; by per-action average, a centre-back and a goalkeeper sit at the foot of the table. A surface built only from skeleton geometry, with no notion of reputation, recovers the game's own hierarchy of creators. It passed.

What are the real limitations?

The sample is five matches, the data released for the challenge. This is a limit on the *input*, not the method: the pipeline is match-agnostic and deterministic, so a full season is a matter of feeding in more data, not changing code. The biomechanical constants are taken from the literature rather than fitted to football tracking; fitting them on a larger sample would sharpen the surface. And the orientation-aware pitch-control function is currently validated indirectly, through DOS.

What did it cost to run?

The metric chain and statistics run on a local 8 GB machine thanks to the memory-safe design. The video renders, being embarrassingly parallel, were produced on AWS compute. No specialised hardware and no GPU are required for the analysis itself.

What is next?

A learned run-detection front-end so DOS scores every run, not only sustained bursts; a direct comparison of the reach-field pitch-control function against a standard model; biomechanical constants fitted on a larger sample; and a latency-optimised build for live deployment.

The full technical write-up, all source code and reproduction instructions are in the repository: github.com/jaime-oriol/Diagonality_3D.